

MODULE 3.4

AAA usage and operation

Network Defense | Cisco Networking Academy

Presented by: Kudzaishe Majeza

Cisco Networking Academy | 2026

Topics:

- What is AAA?
- Authentication
- Authorization
- Accounting Logs
- Summary

What Is AAA?

AAA

AAA stands for:

- **Authentication** – Confirms who the user is.
- **Authorization** – Decides what the user is allowed to do.
- **Accounting** – Records what the user did.

Simple Example

When you log into your university portal:

- You enter your username and password (**Authentication**).
- You can only access your own courses (**Authorization**).
- The system records your login time (**Accounting**).

.

AAA Authentication

Authentication

Authentication checks that a user is really who they claim to be.

Common authentication methods:

- Username and Password
- PIN
- Fingerprint
- Face Recognition
- One-Time Password (OTP)

Example

When you unlock your phone with your fingerprint, the phone first checks your identity before giving you access.



Authorization

Authorization

Authorization decides what a user is allowed to access after they have logged in.

Examples

Administrator

- Install software
- Change settings
- Create users

Standard User

- Open applications
- View files
- Cannot change system settings

Real-Life Example

A bank employee in customer service can view customer information but cannot approve large financial transactions.

AAA Accounting Logs

Accounting

Accounting keeps a record of user activity.

It records things like:

- Login time
- Logout time
- Files accessed
- Commands used
- Failed login attempts

Why is it Important?

- Helps detect suspicious activity
- Supports security investigations
- Helps administrators monitor network usage

Example

If someone tries to log in with the wrong password five times, the accounting logs record those failed attempts.

Summary

Key Points

- Authentication verifies identity.
- Authorization controls what users can access.
- Accounting records user activities.
- Together, AAA improves network security and accountability.